

Quantification of soil macropores under alpine vegetation using computed tomography in the Qinghai Lake Watershed, NE Qinghai–Tibet Plateau

Xia Hu^{a,c,e}, Zong-Chao Li^{a,c}, Xiao-Yan Li^{b,d}, Lian-you Liu^{a,c}

^a Key Laboratory of Environmental Change and Natural Disaster of Ministry of Education, Beijing Normal University, Beijing 100875, China

^b Key Laboratory of Earth Surface Processes and Resource Ecology, Beijing Normal University, Beijing 100875, China

^c Academy of Disaster Reduction and Emergency Management, Beijing Normal University, Beijing 100875, China

^d College of Resources Science and Technology, Beijing Normal University, Beijing 100875, China

^e Corresponding author at: Academy of Disaster Reduction and Emergency Management, Beijing Normal University, No. 19, Xijiekouwai Street, Haidian District, Beijing 100875, China

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ABSTRACT

The importance of soil macropores as preferential pathways for water, air, and chemical movement in soils has long been recognized. However, studies concerning macropores of soils under alpine vegetation remain scarce. The objective of this study was to quantify the architecture of soils beneath Alpine *Kobresia* meadow, *Achnatherum splendens* steppe, and *Potentilla fruticosa* shrubs using computed tomography in the Qinghai Lake Watershed of northeastern Qinghai–Tibet Plateau. Nine soil cores (0–50 cm deep) were taken at three sites with three replicates. At each site, the three cores taken were scanned with a GE HISPEED FX/I Medical Scanner. The number of macropores, macroporosity, and macropore equivalent diameter within the 50-cm soil profile were interpreted from X-ray computed tomography, to analyze soil architecture. The results indicated that soils under *A. splendens* steppe and *P. fruticosa* shrubs had greater macroporosity, and developed deeper and longer macropores than those in the other vegetation types. For the Alpine *Kobresia* meadow, macropores were distributed mainly in the 0–100-mm soil layer, while they were distributed in the 0–200-mm soil layer for *A. splendens* steppe and in the 0–200-mm soil layer for *P. fruticosa* shrub. The large number of macropores found in the soil surface layer under Alpine *Kobresia* meadows can be attributed to “mattic epipedon”. The large number of macropores found in soil under *A. splendens* steppe and *P. fruticosa* shrubs can be attributed to greater root development. Soil hydraulic conductivity was significantly greater for *A. splendens* steppe and *P. fruticosa* shrub (great macroporosity) than for Alpine *Kobresia* meadow (small macroporosity).

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1. Introduction

Soil macropore is one of the important factors of soil structure (Brewer, 1976), which largely control fluid and solute transport, making visualization and quantification of macropore characteristics essential for better understanding and predicting soil hydrological functions. Macropores are large soil voids, often distinct in some manner from the soil matrix, which permit the preferential flow of water and contaminants through the soil profile (Edwin & David, 2009). The importance of macropores as preferential pathways of water, air, and chemicals in the soil has been widely recognized (Beven & Germann, 1982; Lin et al., 2005; Jarvis, 2007). The conductivity of macropores to water flow strongly depends on their 3-D geometry and topology. Macroporosity, the number of macropores, pore length, pore size distribution, continuity, tortuosity, and connectivity are considered significant characteristics that influence water flow and solute

transport through macropores (Perret et al., 2007; Pierret et al., 2002; Bastardie et al., 2003; Peth et al., 2008; Luo et al., 2008). Different types of macropores have distinct geometries and therefore function differently (Lin et al., 1996; Luo et al., 2008). Previous studies showed that soil type and land use were among the main factors influencing pore characteristics (e.g., Gantzer and Anderson, 2002; Zhou et al., 2008; Udawatta et al., 2008; Mooney and Morris, 2008). Luo et al. (2010) reported that soil type, land use, and their interaction, significantly influenced macroporosity, network density, surface area, length density, node density, and mean angle. Moreover, within the same soil type, soils under pasture had greater macroporosity, length density, and node density than those under row crops, especially in the subsoil. Therefore, there would be complex interactions between vegetation and soil structure, and it is necessary to investigate the relationships between soil pore characteristics and soil functions under a wide range of vegetation, soil, and climate conditions.

X-ray computed tomography (CT) has been used in recent years as a new method to quantify soil pores, especially macropore development,

E-mail address: huxia@bnu.edu.cn (X. Hu).

at a much higher resolution than the previous methods, such as dye tracing, spectral image analysis and soil thin section (Grevers et al., 1989; Capowiez et al., 2003; Taina et al., 2008; Munkholm et al., 2012). X-ray CT technique has been proven to be a very powerful and non-destructive technique to visualize and quantify soil structure at different scales (Garbout et al., 2013 and Taina et al., 2008). For example, Garbout et al. (2013) quantified the effects of tillage on topsoil structural quality using X-ray CT, and Katuwal et al. (2015) linked air and water transport in intact soils to macropore characteristics by X-ray CT. Moreover, a few recent studies have used helical medical CT to quantify macropore characteristics and linked them with soil hydraulic functions (Munkholm et al., 2012; 2013). Moreover, Sammartino et al. (2012) proposed a novel method to visualize and characterize preferential flow in undisturbed soil cores using multi-slice helical medical CT. Hu et al. (2015) also quantified the architecture of soils beneath the shrub canopy and in the adjacent interspace grass area during evolution of shrub encroachment using helical medical X-ray computed tomography. These studies suggest that helical medical CT could be viewed as a method suitable for quantifying soil macropores.

Alpine vegetation grows in the high-latitude area and has a unique soil and harsh climate condition, which is characterized by low temperatures, dryness, ultraviolet radiation, freeze and thaw cycle as well as a short growing season. The soil structures under alpine vegetation could be different from that in low-latitude areas; however, studies concerning interactions between soil macropores and alpine vegetation are rare, particularly for the Qinghai–Tibet Plateau. Therefore, the objective of this study was to quantify soil pore features using helical medical CT for soils beneath alpine vegetation in the Qinghai Lake Watershed of the NE Qinghai–Tibet Plateau, and this study would provide a better understanding of the interaction between vegetation types and pore characteristics in the alpine environment.

2. Materials and methods

2.1. Study sites and core samples

This study was conducted in the watershed of the Shaliu River, the second largest river in the Qinghai Lake Watershed, situated in the northeastern of the Qinghai–Tibet Plateau, which lies in the semiarid, cold and high-altitude climate zone. Mean annual temperature and precipitation between 1957 and 2015 were 0.1 °C and 400 mm in the watershed, respectively. Approximately 70%–80% of the annual precipitation occurs in summer and autumn. Winters are clear, dry, and cold, and the summer is warm and wet. The soil types are classified as Typic Cryoboroll and Typic Haploboroll according to USDA taxonomy, or Haplic Kastanozem and Cryi-Haplic Kastanozem according to the World Reference Base for Soil Resources (WRB). The dominant vegetation types are *Kobresia* meadow, *Achnatherum splendens* steppe, and *Potentilla fruticosa* shrub. Alpine meadow is the main ecosystem in the Qinghai Lake Watershed, and it accounts for 55.9% of the total land area (Zhang et al., 2015). Alpine steppe is the second largest ecosystem type, and it occupies 15.6% of the total land area. Alpine shrub meadow is the only ecosystem with a chamaephyte functional group. We selected three sites as sampling areas in each vegetation type (*Kobresia* meadow, *A. splendens* steppe, and *P. fruticosa*) in the Shaliu River watershed (Fig. 1). Landscapes of the three vegetation types are shown in Fig. 2. At Site 1, the main species were *Kobresia pygmaea*, *Saussurea pulchra*, and *Polygonum viviparum*, and the average vegetation cover was $97.00 \pm 2.00\%$. At Site 2, the main species were *A. splendens*, *Artemisia frigid*, *Dracocephalum heterophyllum*, *Heteropappus altaicus*, and *Potentilla saundersiana*, and the *A. splendens* cover was $8.00 \pm 3.00\%$. At Site 3, the main species were *P. fruticosa*, *Carex moorcroftii*, and *P. saundersiana*, and the *P. fruticosa* shrub cover was $21.00 \pm 3.00\%$. A total of 9 undisturbed soil cores were taken under plant canopy at the three sampling sites with three replicates. Plexiglas cylinders with

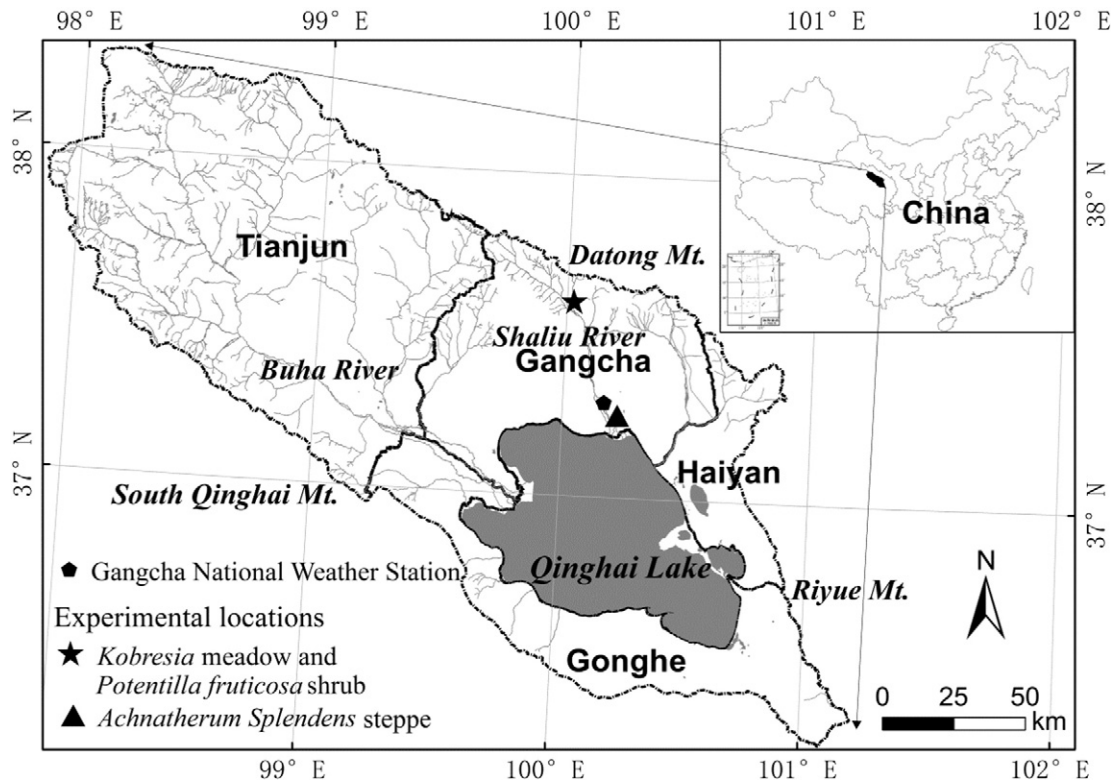


Fig. 1. Study sites and sampling positions.



Fig. 2. Landscapes of the field sites for the Alpine *Kobresia* meadow (A), *Achnatherum splendens* steppe (B), and *Potentilla fruticosa* shrubs (C).

4-mm thick walls, 10 cm diameter and 50 cm length were used to house intact soil. The aboveground shrub or grass was carefully clipped to ground level with minimum disturbance, and litter was completely removed to expose bare soil before soil sampling. We moistened the soil with water before sampling and the core was taken with a polyvinyl chloride (PVC) cylinder with a beveled edge at its bottom. Core extraction followed the procedure of Sammartino et al. (2012). The core was extracted by pushing, slowly and vertically, into the soil, the penetration of the cylinder progressing centimeter by centimeter, carefully removing the soil around the tube bottom at each step, to minimize mechanical stresses during extraction. After sampling, a PVC grid was glued to the base of the cylinder, and the cylinder was then glued to an annular sample holder to protect the bottom of the core from mechanical disturbance. To protect the samples during transport from field to laboratory, the cores were placed in sealed plastic bags, and were wrapped with sponge and packed with wheat straw.

Soil samples for property analysis were collected at 0–50 cm for all the soil types. For grain size distribution, sand fraction (2000 to 50 μm) was measured by sieving, and the silt (50–2 μm) and clay (<2 μm) fractions were determined using the pipette method. Organic carbon content was determined by the method of Walkley and Black (1934) and converted to organic matter content by multiplying by 1.724. The soil characteristics at 0–50 cm under the different vegetation types are listed in Table 1.

2.2. CT scanning and image analysis

A GE HISPEED FX/I Medical Scanner (General Electric Company), a 64-slice spiral CT, was used to scan soil cores at an energy level of 120 kV and 300 mA, with a 0.625-mm scanning interval. This combination provided detailed, low-noise projections, with a thread pitch of 0.0984 and scan speed of 39.38 mm/r. The GE HISPEED FX/I Medical Scanner is an advanced high-speed medical X-ray, 3-D scanning system, developed as an area detector employing flat panel amorphous silicon arrays. After scanning and reorientation, 800 slices were produced for every soil core, in coronal view. Voxel Cone Tracing (VCT) acquired 8-bit and 512×512 images, with a voxel dimension of $0.625 \times 0.33 \times 0.33$ mm (Hu et al., 2015).

Images were analyzed using the Image J 1.48s software to examine the pore characteristics. A 7575-mm² square area was indicated as the Region of Interest with the Region of Interest (ROI) tool, and then the area outside the ROI was deleted to exclude voids near the core walls,

and to minimize beam-hardening interference. A value of 97 (in a range of 0 to 255) was selected as the threshold value for analysis of the images. This value was selected using optimization with known phantoms, which included air-filled tubes. Values lower than the threshold were identified as air-filled pores, and values greater than threshold were identified as non-pore (Feng et al., 2003; Hu et al., 2015).

Thresholding for the binary images was determined by using an artificial macropore. A Plexiglas cylinder with a known diameter was regarded as an artificial macropore and inserted into an undisturbed soil core, then pulled out. The core was then scanned using the Medical CT scanner. First, we assumed one threshold value for the thresholding sample, and then calculated the macropore size based on image analysis using Image J 1.48s computer software, and compared it with the actual size. If the difference between them was too large, we selected another threshold value until the difference was reduced to less than 1% (Li et al., 2013). With a threshold value in Image-J version 1.48s, the images were translated into binary images. In the images, the black areas indicated macropores and the white areas indicated the soil matrix. Three-dimensional visualization of the soil macropore networks in the soil columns was performed using the method of saturated volume rendering using Image J 1.48S. Statistics of individual pores were computed with the Analyze Particles tool. The number of pores and pore area dimension were determined for each image. The macropore area of an image was divided by the 7575-mm² area to estimate macroporosity.

Soil pores were divided into three shape classes according to the ratio of pore area (A) to the square of the pore perimeter (P): values of A/P^2 greater than 0.04 were designated as round, those less than 0.04 and greater than 0.015 as irregular, and those less than 0.015 as elongate (Bouma et al., 1977; Pagliai et al., 1983). For round and irregular pores, the subdivision was determined according to the equivalent pore diameter, $2(A/\pi)^{0.5}$, and for elongate pores it was determined according to pore width, $0.25[P - (P^2 - 16A)]^{0.5}$, which followed the procedures of Pagliai (1983), Pagliai et al. (1983), Valentin (1991), and Gimenez et al. (1992). The macropore equivalent diameter was determined by this method.

2.3. Statistical analyses

Differences among the measured parameters within group sites were analyzed using one-way analysis of variance (ANOVA) and Fisher's protected least significant difference (LSD) test. All

Table 1
Soil properties under three vegetation types.

Vegetation types	Depth	Particle size composition (%)					Organic matter (g kg ⁻¹)
		$\Phi \leq 0.002$ mm	0.002–0.02 mm	0.02–0.05 mm	0.05–0.25 mm	0.25–2.00 mm	
Alpine <i>Kobresia</i> meadow	0–50 cm	8.45	23.13	14.38	13.79	40.26	188.75
<i>Achnatherum splendens</i> steppe	0–50 cm	21.55	27	21.67	27.16	2.3	35.37
<i>Potentilla fruticosa</i> shrub	0–50 cm	14.7	30	22	17.37	15.93	111.38

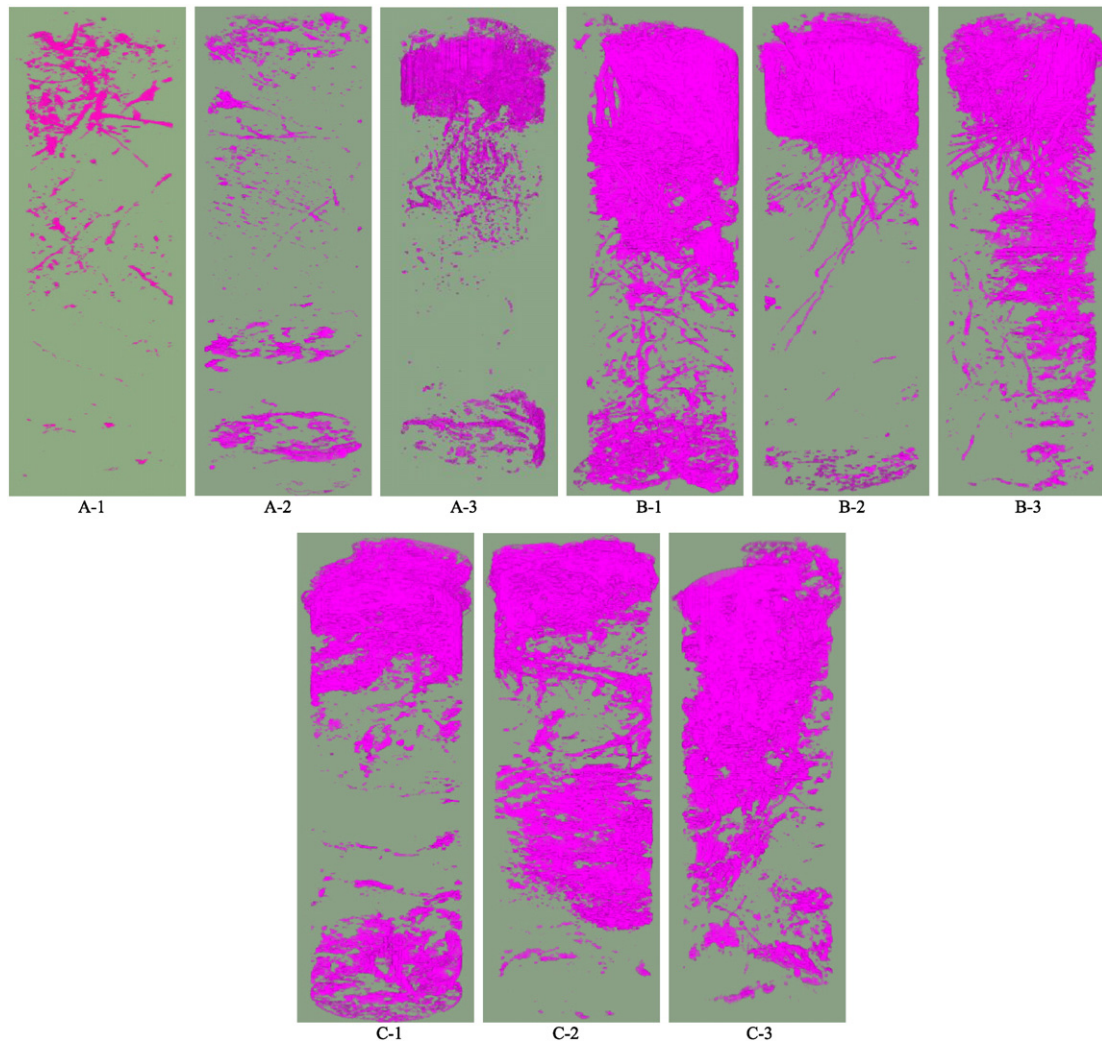


Fig. 3. Three-dimensional visualization of soil macropore networks in the soil columns under Alpine *Kobresia* meadow (A), *Achnatherum splendens* steppe (B), and *Potentilla fruticosa* shrubs (C). The purple is pore and the gray is non-pore.

statistical analyses were conducted using version 13.0 of SPSS software (SPSS Inc., Chicago, IL) at the $p = 0.05$ level of confidence.

3. Results

3.1. Visualization of macropore networks

Three-dimensional visualizations of macropores in the nine soil columns are shown in Fig. 3. It was clear that spatial macropore characteristics were distinctly different among the soils under different

vegetation types. In the vertical soil column, the soil macropores under *A. splendens* steppe and *P. fruticosa* shrubs were greater, deeper, and longer than those under Alpine *Kobresia* meadow. Moreover, large numbers of pores were distributed in the surface soil of Alpine *Kobresia* meadow. In contrast, macropores were distributed throughout the soil profile for both *A. splendens* steppe and *P. fruticosa* shrub. Macropore networks in the soil profile of the *A. splendens* steppe and *P. fruticosa* shrubs were more complex, vertically oriented, and continuous than those of soil of the Alpine *Kobresia* meadow (Fig. 3).

Table 2
Number of CT-measured macropores, macroporosity, and macropores at different soil depths under different vegetation types.

Vegetation types	Soil depth	Macropore number (mean $\pm$ SD)	Mean macropore size (mm)	Macroporosity (%)	Macroporosity (%)				
					0–100 mm	100–200 mm	200–300 mm	300–400 mm	400–500 mm
Alpine <i>Kobresia</i> meadow	0–50 cm	12 (± 5)b	1.72	0.76 (± 0.54)a	2.95 (± 2.20)A	0.22 (± 0.07)	0.05 (± 0.03)	0.05 (± 0.02)	0.59 (± 0.48)
<i>Achnatherum splendens</i> steppe	0–50 cm	29 (± 5)a	1.507	9.10 (± 2.81)c	30.29 (± 6.70)Bb	5.07 (± 4.00)a	0.70 (± 0.38)a	1.18 (± 1.00)a	
<i>Potentilla fruticosa</i> shrub	0–50 cm	33 (± 7)a	1.634	5.52 (± 1.40)b	12.42 (± 1.94)Bb	2.00 (± 1.21)a	2.69 (± 1.61)a	2.35 (± 0.84)a	

Columns for each data group by the same letter are not significantly different at $p = 0.05$; the absence of letters indicates no significant differences for that property. Data in the parentheses are standard deviations.

3.2. CT-measured number of macropores and macroporosity

The number of soil macropores was significantly less for Alpine *Kobresia* meadow than both *A. splendens* steppe and *P. fruticosa* shrubs (Table 2). No significant difference in the number of soil macropores was found between *P. fruticosa* shrubs and *A. splendens* steppe. The soil of *P. fruticosa* shrubs had 33 macropores compared to 29 and 12 for *A. splendens* steppe and Alpine *Kobresia* Meadow, respectively.

There were significant differences in macroporosity among the three vegetation types (Table 2). Soil in the *A. splendens* steppe had 9.10% macroporosity, compared to 5.52% and 0.76% for the *P. fruticosa* shrub and Alpine *Kobresia* meadow, respectively. The magnitude of the macroporosity for the three vegetation types corresponded well with the results for number of macropores.

Macropores of soils were distributed mainly in the shallow soil layer (0–100 mm) for Alpine *Kobresia* Meadow, but in relatively deeper soil for *A. splendens* steppe (0–200 mm) and *P. fruticosa* shrub (0–400 mm). Macroporosity was 2.95, 0.22, 0.05, 0.05, and 0.59%, at soil depths of 0–100, 100–200, 200–300, 300–400, and 400–500 mm, respectively, for the Alpine *Kobresia* Meadow. Soils under *A. splendens* steppe had 30.29, 5.07, 0.70, and 1.18% macroporosity, at soil depths of 0–100, 100–200, 200–300, and 300–400 mm, respectively. The soil under *P. fruticosa* shrubs had 12.42, 2.00, 2.69, and 2.35% macroporosity, at soil depths of 0–100, 100–200, 200–300, and 300–400 mm, respectively. For soil under Alpine *Kobresia* meadow, there were no significant differences in macroporosity at all soil depths, however, macroporosity was significantly higher in the shallowest soil layer (0–100 mm) than in the deeper soil layers (100–200, 200–300, 300–400 mm). For soil under *A. splendens* steppe and *P. fruticosa* shrubs, there were significant differences in macroporosity between the top soil layer (0–100 mm) and the deeper layers.

There were significant differences in macroporosity between soil under Alpine *Kobresia* meadow and soil under *A. splendens* steppe at the shallowest depth (0–100 mm). Macroporosity was significantly higher in soil from *A. splendens* steppe than from Alpine *Kobresia* meadow and *P. fruticosa* shrubs in the shallowest soil layer (0–100 mm) (Table 2). No significant differences in macroporosity were found between soils under *A. splendens* steppe and *P. fruticosa* shrub at the shallowest soil depth. Furthermore, there were no significant differences in macroporosity in soil samples from under Alpine *Kobresia* meadow, *A. splendens* steppe, and *P. fruticosa* shrubs at the deeper soil depths (100–200, 200–300, and 300–400 mm).

To investigate the variation of macroporosity with depth, macroporosity distribution for each sample was calculated slice per slice, i.e. for every 0.625-mm depth interval. The distributions of macroporosity with sample height (profile depth) for soils of the three alpine vegetation types are presented in Fig. 4. For soil under Alpine *Kobresia* meadow, macroporosity was represented by a smooth curve, with a rapid decrease at soil depths from 0 to 100 mm, and relative stability at depths between 100 and 300 mm. In comparison, the macroporosity change with depth was significantly different between the Alpine *Kobresia* meadow and the *A. splendens* steppe. Under *A. splendens* steppe, macroporosity declined sharply at soil depths of 0–200 mm. Macroporosity remained stable at soil depths from 200 to 350 mm, and then increased at 350 mm soil depth.

The general patterns of macroporosity observed in soil from the *A. splendens* steppe and *P. fruticosa* shrubs were similar. Macroporosity declined sharply at soil depths of 0–200 mm for soil under the *P. fruticosa* shrub and under *A. splendens* steppe. It remained stable at soil depths between 200 and 220 mm for the *P. fruticosa* shrubs, and 200 to 350 mm for the *A. splendens* steppe. Macroporosity was greater at 220 mm under *P. fruticosa* shrubs and at 350 mm under the *A. splendens* steppe. For soil under Alpine *Kobresia* meadow, macroporosity decreased quickly in the top 100 mm, and then increased slowly until a depth of about 270 mm. In summary, macropores were distributed mainly in the 0–100-mm soil layer of the Alpine *Kobresia* meadow, in the 0–200-mm soil layer for *A. splendens* steppe, and in the 0–200-mm soil layer under *P. fruticosa* shrubs.

3.3. Macropore size distribution and mean macropore size

The mean macropore equivalent diameter was similar for soils under different vegetation types (Table 2). The mean macropore diameter for soil under Alpine *Kobresia* meadow was the highest, 1.720 mm, compared to 1.634 mm for the *P. fruticosa* shrubs, and 1.507 mm for the *A. splendens* steppe.

For all soils under different vegetation types, the most common macropore diameters were 0–3 mm (Fig. 5). The three size fractions ($0 < D < 1$ mm; $1 < D < 2$ mm; and $2 < D < 3$ mm) accounted for more than 85% of all counted macropores in nearly all horizons for the *A. splendens* steppe and *P. fruticosa* shrub. These results are consistent with previous findings by Iversen et al. (2012), who reported that the most common macropore diameters were 2 and 3 mm, and that these two size fractions accounted for more than 50% of all counted macropores. However, for soil under Alpine *Kobresia* meadow (Fig. 5),

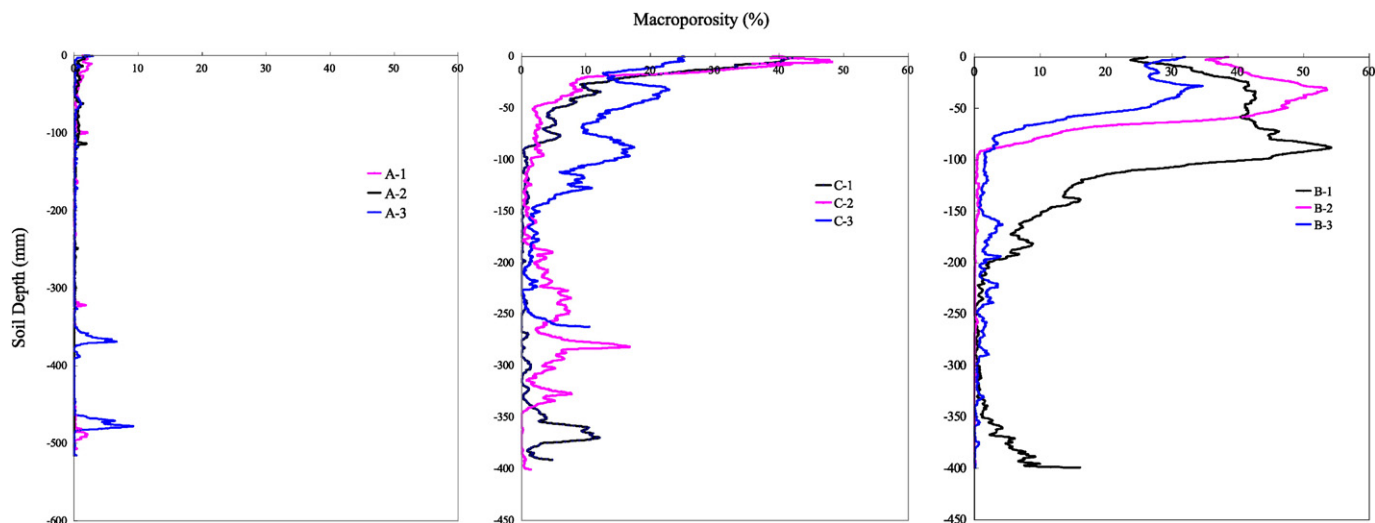


Fig. 4. Distribution of CT-measured macroporosity along the soil column depth under Alpine *Kobresia* meadow (A), *Achnatherum splendens* steppe (B), and *Potentilla fruticosa* shrubs (C).

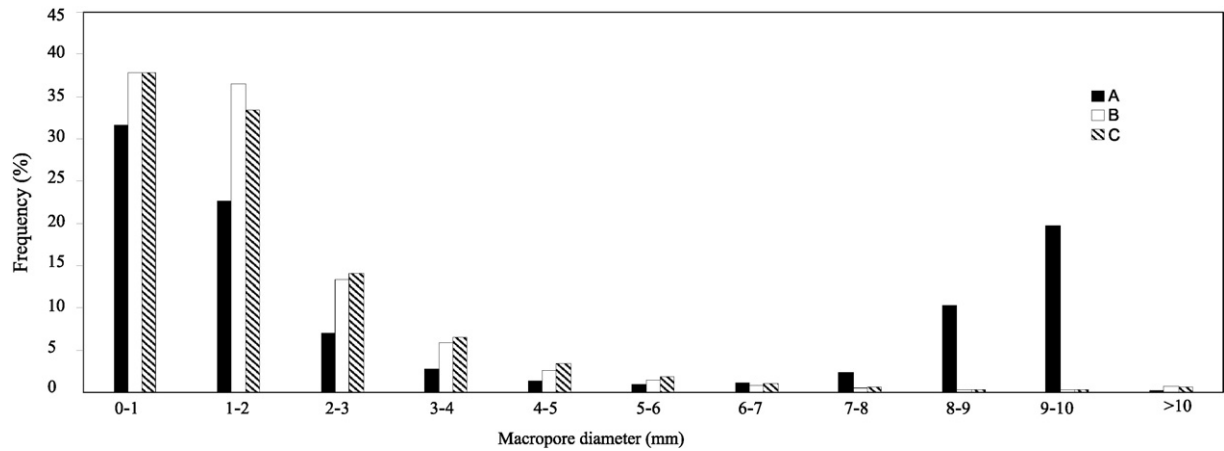


Fig. 5. Distribution of macropores by size for soils under Alpine *Kobresia* meadow (A), *Achnatherum splendens* steppe (B), and *Potentilla fruticosa* shrubs (C).

the most common macropore diameters were 0–1, 1–2, 8–9, and 9–10 mm; and these accounted for more than 85% of all counted macropores.

4. Discussion

Results from this study demonstrated that there were contrasting soil pore characteristics between the three studied alpine vegetation types of *Alpine Kobresia* meadow, *A. splendens* steppe, and *P. fruticosa* shrubs. Soils under *A. splendens* steppe and *P. fruticosa* shrub had greater macroporosity, and developed deeper and longer macropores than those under *Alpine Kobresia* meadow. The spatial distribution of soil macropores corresponded well with the position of roots in the soil profile (Fig. 6). Luo et al. (2010) found that macropores formed by roots were highly continuous and round in shape, their size generally decreasing with depth. For the *A. splendens* steppe, soil horizons were divided into the A (0–20 cm), B (20–60 cm), and C (60–80 cm) layers. The soil

layer with the highest fine-root-biomass density was mainly 0–20 cm (Fig. 6B), and as was the distribution of soil macropores (Fig. 4). Similarly, the soil profile of *P. fruticosa* shrub was characterized by soil horizons of A (0–20 cm), B (20–60 cm), and C (60–85 cm). Plant roots were mostly concentrated in the upper 20-cm soil layer (Fig. 6C), and their macropores were mostly distributed in the 0–200-mm soil layer (Fig. 4). Therefore, the greater macroporosity and more tortuous paths in the *A. splendens* steppe and under *P. fruticosa* shrubs were closely associated with greater root development and subsequent root decay.

In the case of *Alpine Kobresia* meadow, soils developed less macropores compared to those under *P. fruticosa* shrub and *A. splendens* steppe. The reason for macropores in *Alpine Kobresia* meadow being mostly concentrated in the upper 0–100 mm soil layer was mainly attributed to special “mattic epipedon” horizon in the near-surface soil profile (Fig. 6A). The soil horizons of *Alpine Kobresia* meadow consisted of A (mattic epipedon, 0–10 cm), B (10–40 cm), and C (40–90 cm). The A soil layer was a rich surface horizon with intertwined

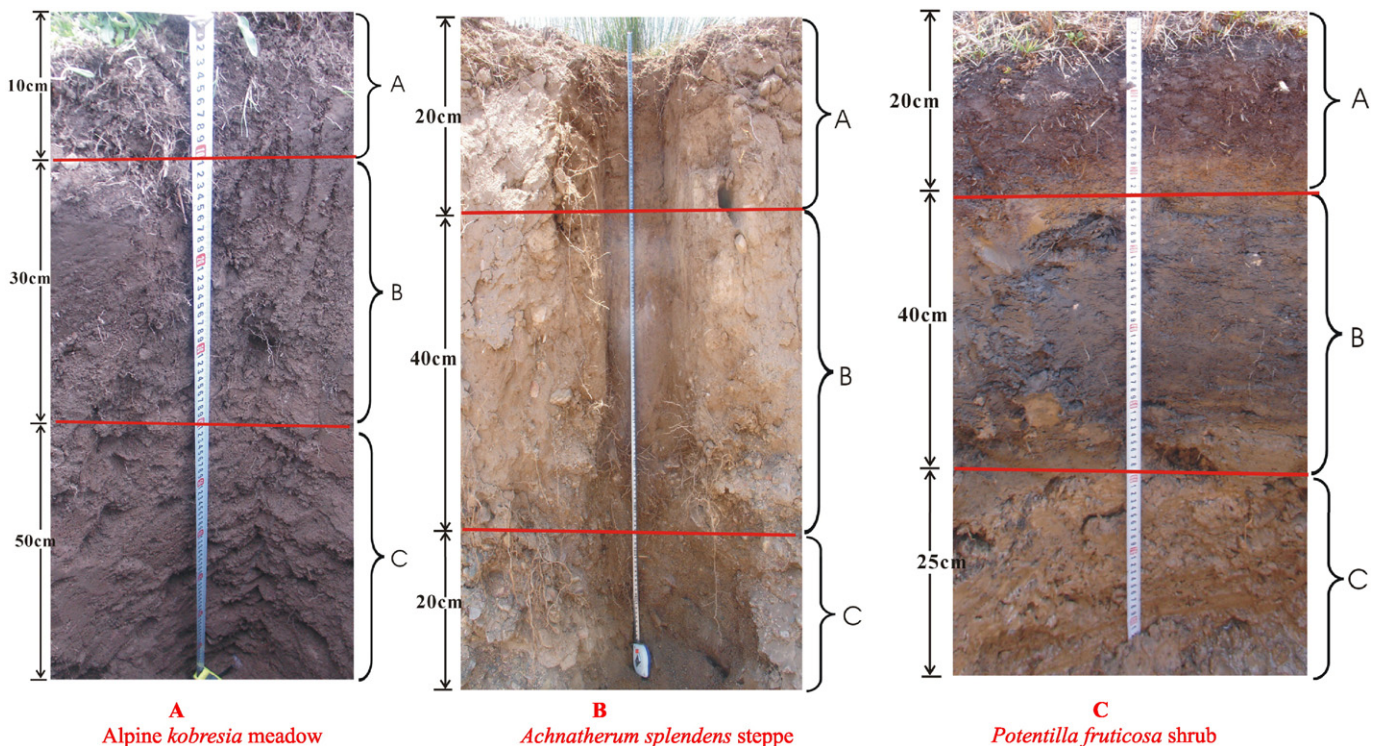


Fig. 6. Soil profiles of different vegetation types (*Alpine Kobresia* meadow, *Achnatherum splendens* steppe, and *Potentilla fruticosa* shrub).

grass roots, which has been recognized in Chinese soil taxonomy as “mattic epipedon” (Fig. 6A) (Gong, 1999; Kaiser et al., 2008; Zeng et al., 2013). Previous studies indicated that the mattic epipedon lowers the saturated soil water content, but increases the soil water storage, and within this layer, plant roots are dense and compressed (Lu et al., 2006; Pang et al., 2009). During the sampling, we had observed in situ that the mattic epipedon was mainly found in the soil surface layer (0–100 mm). In the mattic epipedon, the organic carbon layer had a content of 203 g/kg organic matter in the upper 0–100 mm. The mean organic-matter content in all the soil layers was 188.75 g/kg (Table 1). Organic matter would enhance aeration for biological activities in *Kobresia* meadow soils, which would provide a positive feedback to further macropore development. Moreover, the smaller, more randomly distributed, and less continuously distributed macropores of the Alpine *Kobresia* meadow soil were probably formed by freezing and thawing (Luo et al., 2010). *Kobresia* meadow is situated in the high-altitude area of the Qinghai watershed (3530 m), so freezing and thawing fluctuation occurs frequently in late autumn and early spring. Freezing and thawing processes could increase soil surface-bulk density, but also decrease macroporosity, aggregate stability, and water-storage capacity (Oztas & Fayetorbay, 2003; Wang et al., 2014). Interaction between the dynamics of freezing and thawing, and soil pores under alpine vegetation needs further investigation.

This study preliminarily revealed that great macroporosity could increase soil hydraulic conductivity for soils under alpine vegetation, highlighting the importance of soil macroporosity and its linkage to soil hydrological behavior in the alpine environment (Fig. 7). However, the medium macroporosity appeared to provide the greatest hydraulic conductivity, compared with small and large macroporosity. Because of small number of measurements of hydraulic conductivity, the reason for such pattern was not clear and needs further study. The macropore characteristics roughly corresponded with unsaturated hydraulic conductivity of soils of the three studied vegetation types (Tables 2 and 3, Fig. 7). The average macroporosity was 0.76, 5.52, and 9.10%, for *Kobresia* meadow, *P. fruticosa* shrubs, and *A. splendens* steppe, respectively. The soil unsaturated hydraulic conductivity was significantly greater in *A. splendens* steppe [$K(10) = 0.484 \text{ mm} \cdot \text{h}^{-1}$] and *P. fruticosa* shrub [$K(10) = 0.885 \text{ mm} \cdot \text{h}^{-1}$] than in Alpine *Kobresia* meadow [$K(10) = 0.345 \text{ mm} \cdot \text{h}^{-1}$] (Fig. 7).

Table 3

Unsaturated hydraulic conductivity (mm h^{-1}) of soils under different vegetation types at tensions of 10, 30, and 60 mm.

Vegetation types	$K(10)$	$K(30)$	$K(60)$
Alpine <i>Kobresia</i> meadow	0.345 (± 0.05)a	0.300 (± 0.08)a	0.059 (± 0.01)a
<i>Achnatherum splendens</i> steppe	0.484 (± 0.06)a	0.211 (± 0.02)a	0.076 (± 0.02)a
<i>Potentilla fruticosa</i> shrub	0.885 (± 0.04)b	0.870 (± 0.09)b	0.176 (± 0.02)b

Columns for each data group by the same letter are not significantly different at $p = 0.05$; the absence of letters indicates no significant differences for that property. Data in the parentheses are standard deviations.

5. Conclusions

The purpose of the study was to quantify the architecture of soils beneath Alpine *Kobresia* meadow, *A. splendens* steppe, and *P. fruticosa* shrub using computed tomography in the Qinghai Lake Watershed. The results of the study provided macropore information about the effects of vegetation on soil architecture. Soils under the *A. splendens* steppe and *P. fruticosa* shrubs had more macropores than soils in the Alpine *Kobresia* meadow; moreover, larger numbers of macropores were distributed in the soil surface under Alpine *Kobresia* meadow, while in *A. splendens* steppe, they were found throughout the 0–200-mm soil layer. In the *P. fruticosa* shrubs, macropores were distributed in the 0–200-mm soil layer. Moreover, the soil macropores under *A. splendens* steppe and *P. fruticosa* shrubs were more convoluted and vertically continuous than those under other vegetation types. The large number of macropores found in the soil surface layer under Alpine *Kobresia* meadow could be attributed to the mattic epipedon of Alpine meadow. The large number of macropores found in soil under *A. splendens* steppe and *P. fruticosa* shrub could be attributed to greater root development.

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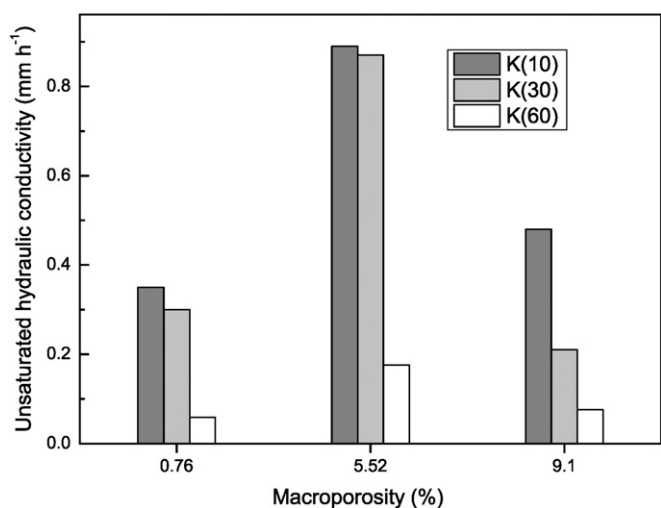


Fig. 7. Relationship between macroporosity and unsaturated hydraulic conductivity (K) for Alpine *Kobresia* meadow, *Achnatherum splendens* steppe, and *Potentilla fruticosa* shrubs.

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